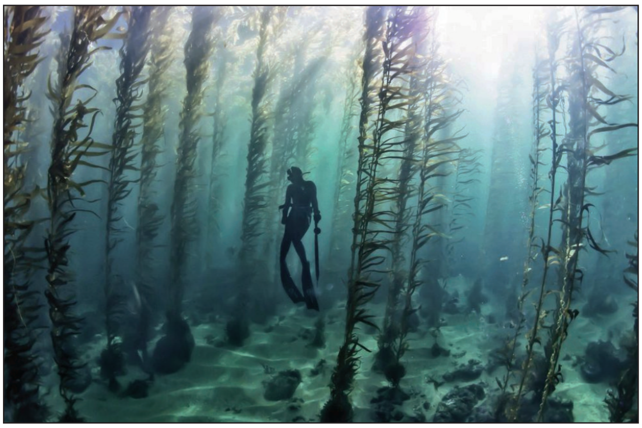
Examiner only

**4.** Seaweeds (marine algae) contain large amounts of iodine in their tissues. The iodine is obtained from the seawater in which they live and is needed to maintain healthy growth of the plant. The iodine content of seawater is about 0.055 mg / kg. Some of the largest seaweeds are the kelps that live in kelp 'forests' in shallow seas and oceans where light can penetrate. Some kelp species are over 80 m in length and can grow 0.5 m in a day.

Photo showing part of a kelp 'forest' with sub-aqua diver



*(a)* State the reason why kelp only grows in seawater where sunlight can penetrate. [1]

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*(b)* Kelp is of commercial importance. It is harvested for many reasons including as a source of iodine and other chemicals for the pharmaceutical industry. Conservationists are concerned that in some parts of the world kelp harvesting is reducing the number of other organisms living in the kelp forests.

Harvesting kelp in Canada



 

Examiner only

Using **only the information given**, what could you, as a scientist, say to the conservationists that would help them understand that a certain level of kelp harvesting could be maintained? [1]

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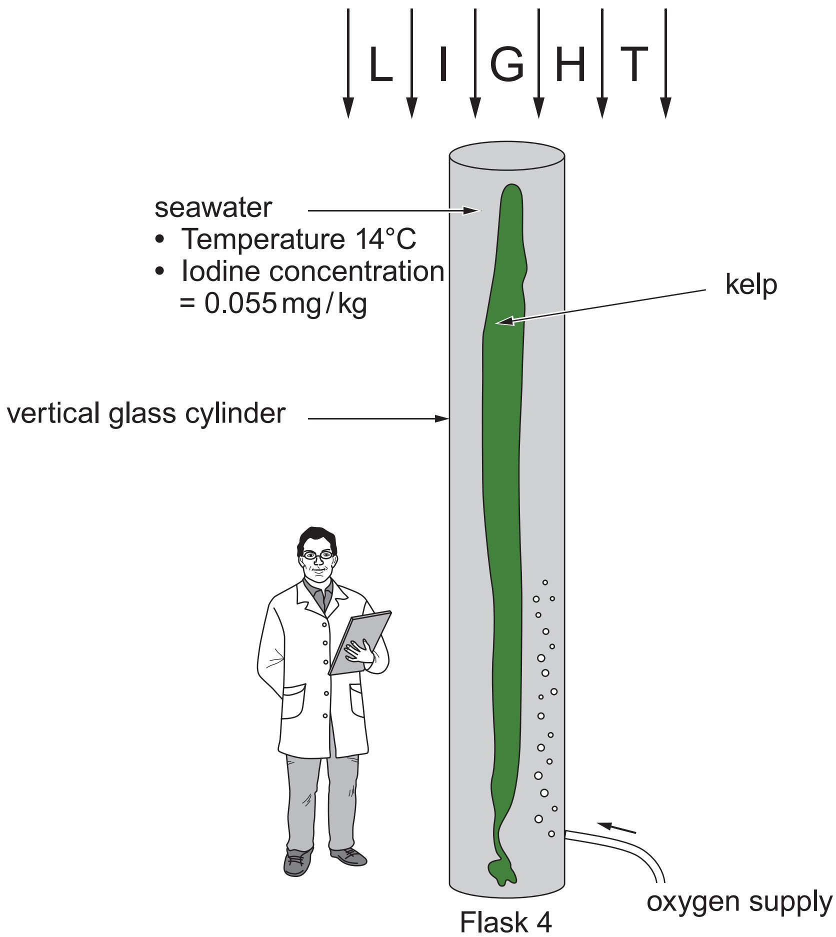
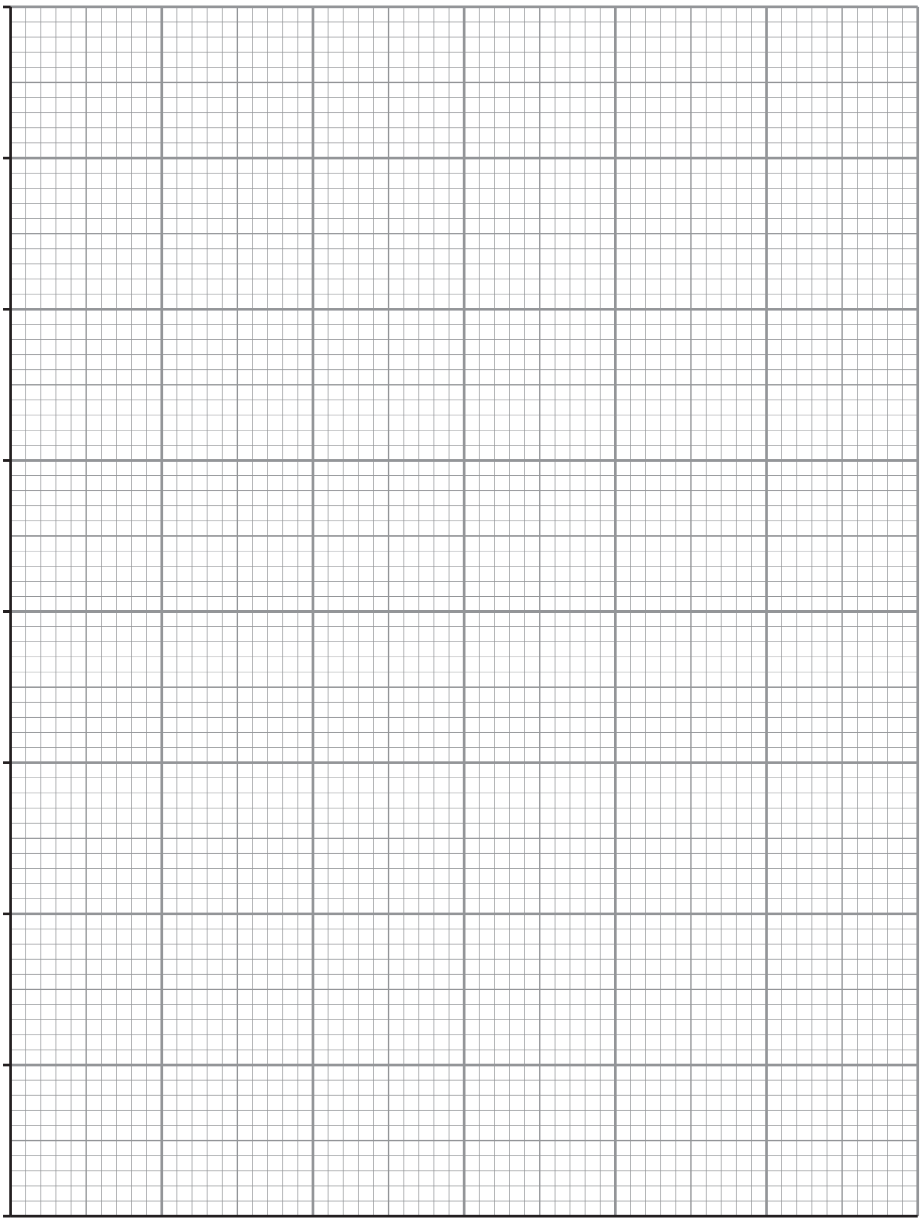
(c) The increasing importance of the commercial use of kelp has led scientists to conduct experiments in an attempt to grow them in land based factories. In one experiment sugar kelp *(Laminaria saccharina)* was grown in seawater in large vertical glass flasks. Scientists were trying to establish if the concentration of oxygen contained in the seawater affected the absorption of iodine by the kelp.




One set of results from the experiment is shown in the table.

| Flask No. | Rate of flow of oxygen into flask ($dm^3$/minute) | Mass of iodine extracted from kelp (mg /kg) |
|---|---|---|
| 1 | 0.0 | 45 |
| 2 | 0.5 | 140 |
| 3 | 1.0 | 259 |
| 4 | 1.5 | 676 |
| 5 | 2.0 | 740 |
| 6 | 2.5 | 780 |
| 7 | 3.0 | 780 |

(i) On the grid below plot a line graph for the mass of iodine against the rate of oxygen flow. You must add suitable scales to each axis. Join the plots with a ruler. [4]



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Examiner only

(ii) Describe the effect of increasing the rate of flow of oxygen above 2.5 $dm^3$/minute on the mass of iodine extracted. [1]

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(iii) The kelp used in the 7 flasks were of different sizes but the experimental results could still be compared. Explain how this was possible. [1]

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(iv) For **Flask 4** calculate how many times greater the concentration of iodine is in the kelp compared to the concentration of iodine in seawater. **Give your answer in standard form.** [2]

Answer = .............................................. × greater

(v) The only source of iodine for kelp is the seawater in which they live. Explain how kelp is able to accumulate iodine against a concentration gradient. [3]

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*(d)* Another group of scientists want to test the reproducibility of the above experiment. State **one** *other* controlled variable, the value of which they would need to know, before they could proceed. [1]

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